Monopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 16, Monopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 16, "Monopoly".

What you must know from this chapter

1. Definition of a monopoly.
2. How can a monopoly arise?
3. Difference between P and MR.
4. Graphical representation of profits/losses.
5. Barriers to entry.
6. Comparison to perfect competition.
7. Natural monopoly.

Part 1

Monopoly and Market Power

Monopoly

Definition: Monopoly
A firm that is the sole seller of a product without close substitutes.

Definition: Market power
The ability of a firm to influence the market price of what it sells.

- A competitive firm is a price taker. A monopoly is a price maker.

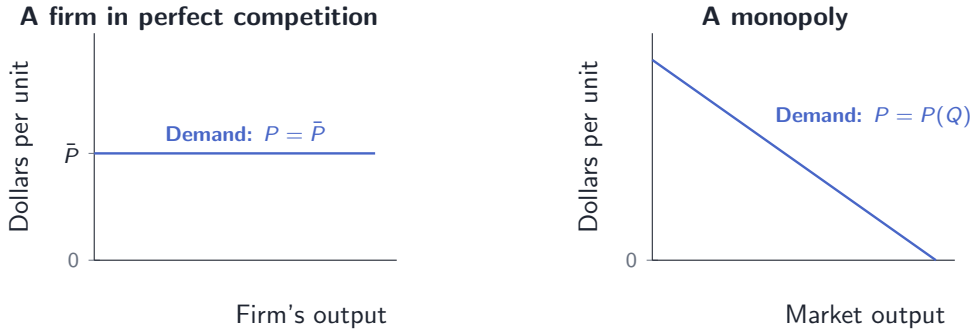
A price maker faces the market demand curve

- A competitive firm sells as many units as it likes at the market price: price is parametric, $P = \bar{P}$.
- A monopoly is the whole market. It faces the market demand curve, so its price depends on its output: $P = P(Q)$.
- Demand slopes down. If the monopoly wants to sell more, it must lower its price.
- It chooses Q; the demand curve then gives the highest price buyers will pay, $P(Q)$.

Finding

A monopoly cannot charge any price and sell any quantity. It can only choose a point on its demand curve.

Competitive firm versus monopoly



Finding

A competitive firm faces a horizontal demand line at $\bar{P}$. A monopoly faces the downward-sloping market demand curve, so selling more means a lower price.

Example: a cartridge seller

One firm is the only seller of a printer cartridge. Its demand curve:

$$P(Q) = 32 - 2Q.$$

Each extra cartridge sold lowers the price by \$2.

To sell 6 cartridges, the most it can charge is \$20. To charge \$24, it can sell only 4.

Q	$P(Q) = 32 - 2Q$
2	\$28
4	\$24
6	\$20
8	\$16

Part 1 summary

- A monopoly is the sole seller of a product without close substitutes.
- It is a price maker: it faces the market demand curve, $P = P(Q)$.
- To sell more it must lower its price. It can only choose points on its demand curve.

Your turn: Check your understanding

The cartridge seller's demand is $P(Q) = 32 - 2Q$.

1. At 4 cartridges, the price is:
   A. \$16 B. \$24 C. \$28
2. To charge \$20, it can sell:
   A. 4 B. 6 C. 10
3. A monopoly can charge any price and still sell any quantity.
   A. True B. False

Q	$P(Q)$
2	\$28
4	\$24
6	\$20
8	\$16

Solution: Check your understanding

1. **B.** $P(4) = 32 - 2 \times 4 = \$24$.
2. **B.** $20 = 32 - 2Q$, so $Q = 6$.
3. **B.** False: it can only choose points on its demand curve.

Q	$P(Q)$
2	\$28
4	\$24
6	\$20
8	\$16

Check: Part 1

Monopoly firms face

A. downward-sloping demand curves, so they can sell as much output as they desire at the market price.

B. downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.

C. horizontal demand curves, so they can sell as much output as they desire at the market price.

D. horizontal demand curves, so they can sell only a limited quantity of output at each price.

Check: Part 1 – answer

Monopoly firms face

A downward-sloping demand curves, so they can sell as much output as they desire at the market price.

B downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.

C horizontal demand curves, so they can sell as much output as they desire at the market price.

D horizontal demand curves, so they can sell only a limited quantity of output at each price.

Finding

A monopoly is the whole market, so it faces the downward-sloping market demand curve. Each output has one highest price, $P(Q)$. Horizontal demand describes a competitive firm.

Part 2

Barriers to Entry and Natural Monopoly

Barriers to entry

Definition: Barrier to entry
Anything that stops other firms entering a market and competing.

Without barriers, profit attracts entry and the monopoly disappears. Monopolies arise from three sources:

1. **A key resource** is owned by one firm (a unique mineral deposit).
2. **The government** gives one firm the exclusive right to sell (a patent or a copyright).
3. **Costs:** one firm can supply the whole market more cheaply than two or more firms. This is a natural monopoly.

Natural monopoly

Definition: Natural monopoly

A market in which one firm can supply the whole market at a lower average total cost than two or more firms could.

- It arises when fixed cost FC is large and marginal cost c is small and constant: $\text{TC} = \text{FC} + c \times Q$.
- Then $\text{ATC} = \dfrac{\text{FC}}{Q} + c$. As Q rises, FC is spread over more units, so ATC falls.

Finding

If ATC falls over the whole market, splitting output between two firms raises ATC: each firm pays the fixed cost.

Example: a pipeline network

Building the network costs \$120 (FC). Each connection costs \$3 (MC):

$$\text{TC} = 120 + 3Q, \qquad \text{ATC} = \frac{120}{Q} + 3.$$

ATC falls as connections rise: \$9 at 20, \$5 at 60.

ATC is always above MC, which is \$3.

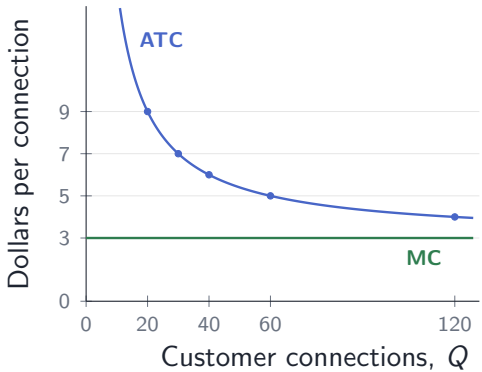
Q	TC	ATC
20	\$180	\$9
30	\$210	\$7
40	\$240	\$6
60	\$300	\$5

Natural monopoly on the graph

$\text{ATC} = 120/Q + 3$ falls over the whole range.

MC is flat at \$3, below ATC at every output.

The more connections one network serves, the lower its cost per connection.



One network or two?

The market needs 60 connections.

- One network: $\text{TC} = 120 + 3 \times 60 = \$300$. $\text{ATC} = \$5$.
- Two networks of 30 each: $2 \times (120 + 3 \times 30) = 2 \times 210 = \$420$. $\text{ATC} = \$7$.
- Two networks cost $420 - 300 = \$120$ more: the second fixed cost.

Finding

In a natural monopoly, one firm supplies the market at the lowest cost. A second firm would duplicate the fixed cost.

Part 2 summary

- Barriers to entry are the cause of monopoly: without them, entry competes profit away.
- Three sources: a key resource, government-created rights, and costs.
- Natural monopoly: large FC and low MC make ATC fall as output rises, so one firm supplies the market most cheaply.

Your turn: Check your understanding

The pipeline network: $\text{TC} = 120 + 3Q$.

1. ATC at 40 connections is:

   A. \$3 B. \$6 C. \$240

2. Serving 40 connections with two networks of 20 instead of one costs extra:

   A. \$0 B. \$60 C. \$120

3. A patent creates a barrier to entry.

   A. True B. False

Q	TC	ATC
20	\$180	\$9
40	\$240	\$6

Solution: Check your understanding

1. **B.** ATC $= 120/40 + 3 = 3 + 3 = \$6$.
2. **C.** Two networks: $2 \times 180 = \$360$; one: \$240. Extra: $360 - 240 = \$120$.
3. **A.** True: the government gives one firm the exclusive right to sell.

Q	TC	ATC
20	\$180	\$9
40	\$240	\$6

Check: Part 2

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

D a larger number of firms will lead to a higher average total cost.

Check: Part 2 – answer

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

D a larger number of firms will lead to a higher average total cost.

Finding

ATC falls as one firm's output rises. Splitting output among more firms means each pays the fixed cost and produces less, so ATC is higher: the network's \$5 becomes \$7 with two firms.

Part 3

Total, Average and Marginal Revenue

Total, average and marginal revenue

Definition: Total revenue

$\text{TR}(Q) = P(Q) \times Q$: the price depends on output.

Definition: Average revenue

$\text{AR} = \text{TR}/Q = P(Q)$: revenue per unit is the price, as in competition.

Definition: Marginal revenue

$\text{MR} = \Delta\text{TR}/\Delta Q$: the extra revenue from an extra unit. $\Delta Q = 1$.

Selling more changes revenue in two ways

Total revenue is $\text{TR} = P \times Q$. Raise output by ΔQ. To sell the extra units, the price must change by ΔP (a fall, so $\Delta P < 0$).

- **Output effect:** the extra ΔQ units sell at price P: revenue rises by $P \times \Delta Q$.
- **Price effect:** the Q units already sold now sell at a lower price: revenue changes by $Q \times \Delta P$, which is negative.
- Together:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

Finding

P is the new price; Q is the number of units already sold.

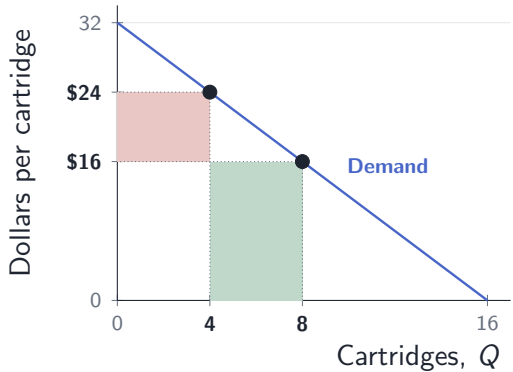
The two effects on the graph

The cartridge seller goes from 4 to 8 units. The price falls from \$24 to \$16: $\Delta Q = 4$, $\Delta P = -8$.

Output effect (green): 4 more units at \$16: $16 \times 4 = +\$64$.

Price effect (red): the first 4 units lose \$8 each: $4 \times (-8) = -\$32$.

$\Delta \text{TR} = 64 - 32 = \$32$. Check: TR goes from \$96 to \$128.



From the change in revenue to marginal revenue

- Start from the two effects:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

- Divide both sides by ΔQ:

$$\frac{\Delta \text{TR}}{\Delta Q} = P \times \frac{\Delta Q}{\Delta Q} + Q \times \frac{\Delta P}{\Delta Q}.$$

- $\Delta Q / \Delta Q = 1$ and $\Delta \text{TR} / \Delta Q = \text{MR}$, so

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}.$$

Marginal revenue is below price (= average revenue)

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}$$

- Demand slopes down, so $\Delta P/\Delta Q < 0$. So the last term is negative.
- Write it as $-k$, where $k > 0$. Then $\text{MR} = P - k$, so

$$P = \text{MR} + k.$$

- Whatever MR is, P is bigger by k. So $P > \text{MR}$.
- $\text{AR} = P$, so also $\text{AR} > \text{MR}$.

MR for a straight-line demand curve

Demand is a straight line: $P = a - bQ$, with $a > 0$ and $b > 0$. Each extra unit lowers the price by b, so $\Delta P / \Delta Q = -b$.

$$
\begin{aligned}
\text{MR} &= P + Q \times \frac{\Delta P}{\Delta Q} \\
&= (a - bQ) + Q \times \frac{\Delta P}{\Delta Q} && \text{replace } P \text{ with } a - bQ \\
&= (a - bQ) + Q \times (-b) && \text{replace } \Delta P / \Delta Q \text{ with } -b \\
&= a - bQ - bQ && Q \times (-b) = -bQ \\
&= a - 2bQ && -bQ - bQ = -2bQ
\end{aligned}
$$

MR is twice as steep as demand

- Demand: $P = a - bQ$. Each extra unit lowers the price by b.
- Marginal revenue: $\text{MR} = a - 2bQ$. Each extra unit lowers MR by $2b$.
- Both lines start at a on the vertical axis.

Finding

For every straight-line demand curve, the MR line starts at the same point and is twice as steep. This is a general result, not a feature of one example.

Example: the cartridge seller's revenue

Q	$P(Q)$	TR	MR
1	\$30	\$30	\$30
2	\$28	\$56	\$26
3	\$26	\$78	\$22
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10
7	\$18	\$126	\$6
8	\$16	\$128	\$2
9	\$14	\$126	−\$2

$P(Q) = 32 - 2Q$; $\text{TR} = P(Q) \times Q$; MR in row Q is $\text{TR}(Q) - \text{TR}(Q-1)$.

MR is below the price for every unit after the first.

The 9th unit has negative MR: TR falls from \$128 to \$126.

The price falls \$2 per unit; MR falls \$4 per unit: twice as steep.

The two effects in the table

Cartridges: $\Delta P/\Delta Q = -2$. Use $\text{MR} = P + Q \times \Delta P/\Delta Q$, with P the new price and Q the units already sold.

- **6th unit:** $\text{MR} = 20 + 5 \times (-2) = 20 - 10 = \$10$.
- **9th unit:** $\text{MR} = 14 + 8 \times (-2) = 14 - 16 = -\$2$.

Finding

When the price effect is larger than the output effect, MR is negative: selling more lowers total revenue.

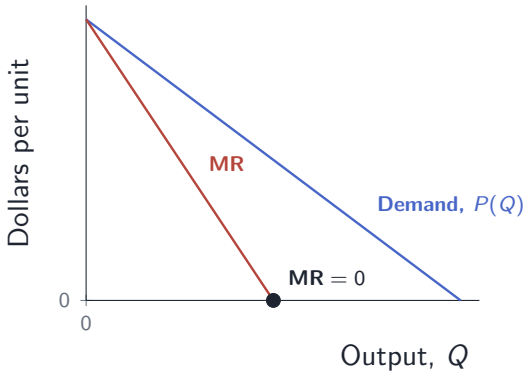
Demand and marginal revenue on the graph

Demand shows the price, $P(Q)$, at each output.

MR lies below demand: MR $< P$.

For a straight-line demand curve, MR starts at the same point and falls twice as fast.

Where MR $= 0$, total revenue is at its largest.



Part 3 summary

- $\text{TR}(Q) = P(Q) \times Q$ and $\text{AR} = P(Q)$.
- $\Delta\text{TR} = P \times \Delta Q + Q \times \Delta P$: an output effect $(+)$ and a price effect $(-)$.
- $\text{MR} = P + Q \times \Delta P/\Delta Q$. Monopoly: $\Delta P/\Delta Q < 0$, so $\text{MR} < P$. Competition: $\text{MR} = \bar{P}$.
- MR can be negative: then selling more lowers total revenue.

Your turn: Check your understanding

The cartridge seller, $P(Q) = 32 - 2Q$.

1. TR at 4 cartridges is:
   A. \$24 B. \$78 C. \$96
2. MR of the 4th cartridge is:
   A. \$18 B. \$24 C. \$96
3. For the 4th cartridge, the price effect changes revenue by:
   A. $-\$2$ B. $-\$6$ C. $-\$24$

Q	$P(Q)$	TR
3	\$26	\$78
4	\$24	\$96

Solution: Check your understanding

1. **C.** $\text{TR} = 24 \times 4 = \$96$.
2. **A.** $\text{MR} = 96 - 78 = \$18$.
3. **B.** $Q \times \Delta P = 3 \times (-2) = -\$6$. Check: $24 - 6 = \$18$.

Q	$P(Q)$	TR
3	\$26	\$78
4	\$24	\$96

Check: Part 3

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

C marginal revenue is less than the price of the product.

D marginal revenue is greater than the price of the product.

Check: Part 3 – answer

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

C marginal revenue is less than the price of the product.

D marginal revenue is greater than the price of the product.

Finding

To sell one more unit, the monopoly lowers the price on all its units. The price effect makes MR $< P$. Average revenue always equals the price, so A and B are wrong.

Part 4

Profit Maximization and Profit or Loss

The rule carries over from competition

- Profit depends on output: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
- One more unit changes profit by $\Delta\pi = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, increase output. If $\text{MC} > \text{MR}$, cut output.
- Profit is maximized where $\text{MR} = \text{MC}$.

Finding

The rule is the same as in competition. The difference: for a monopoly $P > \text{MR}$, so at the best output $P > \text{MC}$.

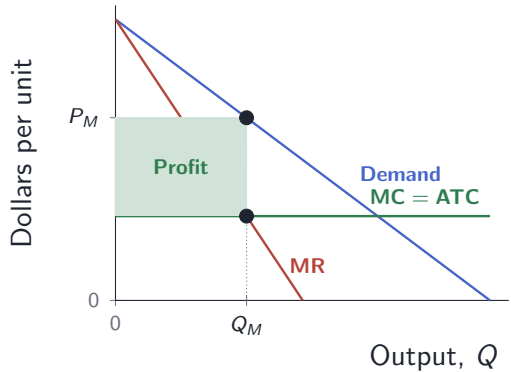
Three steps for a monopoly

Step 1. Output Q_M where MR = MC.

Step 2. Go up to the demand curve: the price is $P_M = P(Q_M)$.

Step 3. Profit $= (P_M - \text{ATC}) \times Q_M$.

The price is set by demand, not by MC: $P_M >$ MC.



Example: the cartridge seller's best output

Q	$P(Q)$	TR	MR	MC	Profit
4	\$24	\$96	\$18	\$10	\$56
5	\$22	\$110	\$14	\$10	\$60
6	\$20	\$120	\$10	\$10	\$60
7	\$18	\$126	\$6	\$10	\$56
8	\$16	\$128	\$2	\$10	\$48

Each cartridge costs \$10 to make. No fixed cost, so ATC $=$ MC $=$ \$10 and profit $=$ TR $- 10Q$.

Unit 6: MR $=$ MC $=$ \$10, so $\Delta\pi = 0$. Best output 6.

Price from demand: $P(6) =$ \$20, above MC $=$ \$10.

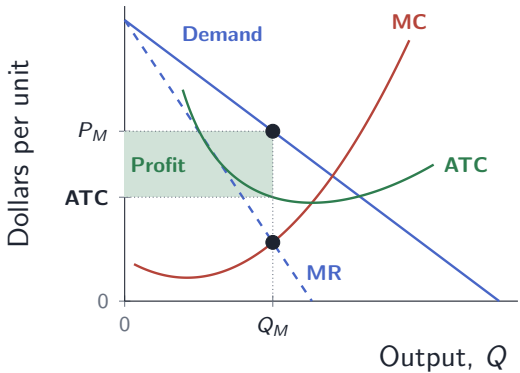
Profit $= (20 - 10) \times 6 =$ \$60.

Profit on the graph

A general monopoly with U-shaped costs.

Choose Q_M where MR = MC. Read P_M from demand and ATC at Q_M.

$P_M >$ ATC: profit. Rectangle height $P_M -$ ATC, width Q_M.



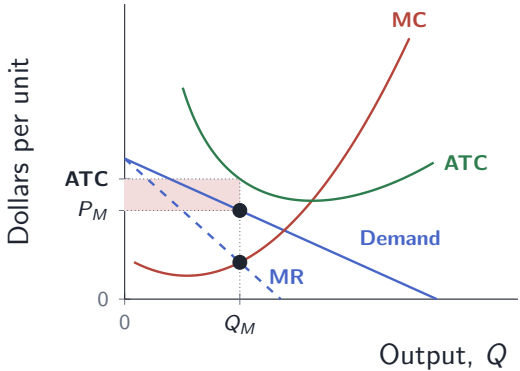
Loss on the graph

The same costs, weaker demand.

Q_M where MR = MC; P_M from demand.

$P_M <$ ATC: a loss. Rectangle height ATC $- P_M$, width Q_M.

Being a monopoly does not guarantee a profit.



A monopoly has no supply curve

- A supply curve gives the quantity a firm offers at each price it takes as given.
- A monopoly does not take the price as given. It chooses output and price together from MR, MC and demand.
- The same MC can go with different prices, depending on the shape of demand.

Finding

A competitive firm's supply curve is its MC curve above minimum AVC. A monopoly has no supply curve: its output depends on demand.

Part 4 summary

- Profit is maximized where MR = MC: $\Delta\pi = \text{MR} - \text{MC} = 0$.
- The price comes from the demand curve at that output: $P_M = P(Q_M)$.
- Because $P >$ MR, the monopoly price is above marginal cost.
- Profit $= (P_M - \text{ATC}) \times Q_M$. If $P_M <$ ATC, the monopoly makes a loss.
- A monopoly has no supply curve.

Your turn: Check your understanding

The cartridge seller's cost rises to \$14 a cartridge (no fixed cost).

1. The best output is:
   A. 4 B. 5 C. 9
2. The price is:
   A. \$14 B. \$22 C. \$24
3. The profit is:
   A. \$40 B. \$70 C. \$110

Q	$P(Q)$	TR	MR
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10

Solution: Check your understanding

1. **B.** Unit 5: MR $= \$14 =$ MC.
2. **B.** From demand: $P(5) = 32 - 10 = \$22$.
3. **A.** Profit $= (22 - 14) \times 5 = \$40$.

Q	$P(Q)$	TR	MR
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10

Check: Part 4

Kayla's Seitan Shop is the only place within 100 miles that sells seitan Buffalo-style wings. Assuming that Kayla is a monopolist and maximizing her profit, which of the following statements is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

B The price of Kayla's seitan wings will exceed Kayla's marginal cost.

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

Check: Part 4 – answer

A profit-maximizing monopolist: which statement is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

B The price of Kayla's seitan wings will exceed Kayla's marginal cost.

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

Finding

Kayla produces where MR $=$ MC and charges the price on her demand curve. For a monopoly $P >$ MR, so $P >$ MC. Costs matter: MC sets her output.

Part 5

Monopoly versus Perfect Competition

Two output rules

- Both firms produce where MR $=$ MC.
- Competition: MR $= \bar{P}$, so $P =$ MC.
- Monopoly: MR $< P$, so $P >$ MC.
- With the same demand and costs, the monopoly stops where MR, not price, meets MC: a smaller output at a higher price.

Finding

Monopoly: lower quantity, higher price. Competition: output where $P =$ MC.

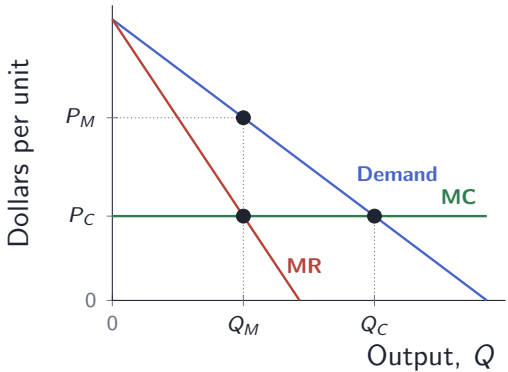
Monopoly and competition on one graph

Same demand and constant MC.

Competition: Q_C where demand meets MC; price $P_C = \text{MC}$.

Monopoly: Q_M where MR meets MC; price P_M from demand.

$Q_M < Q_C$ and $P_M > P_C$.



Example: cartridges under monopoly and competition

Demand $P(Q) = 32 - 2Q$; each cartridge costs \$10.

- Monopoly (Part 4): MR = MC at 6 cartridges, price \$20.
- Competition: $P = \text{MC}$, so $32 - 2Q = 10$ and $Q = 11$, price \$10.
- The monopoly sells 5 fewer cartridges at a price \$10 higher.

	Q	P	$P - \text{MC}$
Monopoly	6	\$20	\$10
Competition	11	\$10	\$0

The deadweight loss of monopoly

Definition: Deadweight loss

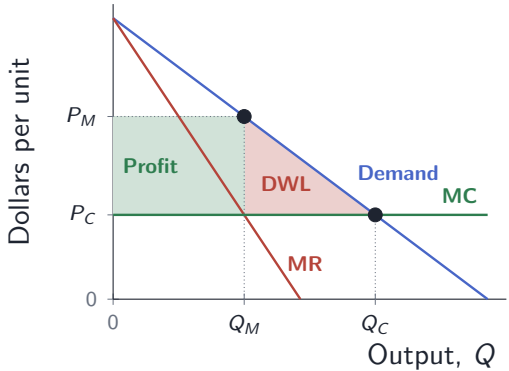
The fall in total surplus when output is below the efficient level.

- The efficient output is where $P = \text{MC}$: buyers' value of the last unit equals its cost.
- The monopoly stops at $Q_M < Q_C$. For each unit between them, buyers' value (demand) is above its cost (MC), but the unit is not made.
- Cartridges: the triangle between demand and MC, $\frac{1}{2} \times (11 - 6) \times (20 - 10) = \$25$.

The deadweight loss on the graph

The triangle between demand and MC, from Q_M to Q_C, is the deadweight loss.

With no fixed cost, ATC = MC, so the profit rectangle is $(P_M - \text{MC}) \times Q_M$. It is not lost: it moves from buyers to the firm.



Finding

Monopoly profit is a transfer from buyers to the firm. The deadweight loss is surplus that no one gets: the social cost of monopoly.

The two market structures

	Perfect competition	Monopoly
Number of firms	Many	One
Product	Identical	No close substitutes
Entry	Free	Blocked by barriers
Demand facing the firm	Horizontal at $\bar{P}$	Market demand, $P(Q)$
Price and MR	$P = \text{MR}$	$P > \text{MR}$
Output rule	$P = \text{MC}$	$\text{MR} = \text{MC}$, $P > \text{MC}$
Long-run profit	Zero	Can stay positive
Deadweight loss	None	Yes

Part 5 summary

- Competition produces where $P = \text{MC}$; monopoly where $\text{MR} = \text{MC}$, with $P > \text{MC}$.
- With the same demand and costs, monopoly output is lower and its price higher.
- The units between Q_M and Q_C are worth more to buyers than they cost: their lost surplus is the deadweight loss, $\frac{1}{2} \times (Q_C - Q_M) \times (P_M - \text{MC})$ for straight-line demand and constant MC.
- Monopoly profit is a transfer, not a deadweight loss.

Your turn: Check your understanding

Cartridges: $P(Q) = 32 - 2Q$, cost \$14 a cartridge.
The monopoly sells 5 at \$22 (Part 4).

1. The competitive output is:
   A. 5 B. 9 C. 14
2. The monopoly price exceeds MC by:
   A. \$0 B. \$8 C. \$22
3. The deadweight-loss triangle is:
   A. \$16 B. \$32 C. \$40

	Q	P
Monopoly	5	\$22
Competition	?	\$14

Solution: Check your understanding

1. **B.** $P = \text{MC}$: $32 - 2Q = 14$, so $Q = 9$.
2. **B.** $22 - 14 = \$8$.
3. **A.** Triangle:
   $\frac{1}{2} \times (9 - 5) \times (22 - 14) = \frac{1}{2} \times 4 \times 8 = \$16$.

	Q	P
Monopoly	5	\$22
Competition	9	\$14

Check: Part 5

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

B above marginal cost.

C equal to demand.

D above demand.

Check: Part 5 – answer

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

B above marginal cost.

C equal to demand.

D above demand.

Finding

With $P >$ MC, some buyers value extra units above their cost, but the monopoly does not produce them. That lost surplus is the deadweight loss. Efficiency needs $P =$ MC.

Part 6

Price Discrimination and Public Policy

Price discrimination

Definition: Price discrimination

Selling the same good at different prices to different buyers.

It needs market power and buyers who cannot resell.

One price, \$70: 3 sold, profit $3 \times 60 = \$180$. (At \$30: $7 \times 20 = \$140$.)

Two prices: $3 \times 60 + 4 \times 20 = \$260$. Output rises from 3 to 7.

Buyers	Number	Most each pays
Large firms	3	\$70
Small firms	4	\$30
Each audit costs \$10.		

Finding

Price discrimination raises profit and can raise output, which cuts the deadweight loss.

Perfect price discrimination

- The monopoly charges each buyer exactly the most that buyer will pay.
- Each extra sale no longer lowers the price on other units, so MR $= P$.
- The monopoly produces where $P =$ MC: the efficient output.

Finding

Perfect price discrimination removes the deadweight loss, but all of the surplus goes to the firm as profit. Buyers get none.

Public policy toward monopolies

1. **Antitrust law:** block mergers and break up firms that reduce competition.
2. **Regulation:** set the price the monopoly may charge.
3. **Public ownership:** the government runs the monopoly.
4. **Do nothing:** when the cure costs more than the deadweight loss.

Finding

Natural monopoly is the hard case: breaking it up would raise costs, so the usual choice is regulation or public ownership.

Regulating a natural monopoly

The pipeline network: $\text{TC} = 120 + 3Q$, $\text{MC} = \$3$. At 60 connections, $\text{ATC} = \$5$.

- **Price = MC = \$3:** efficient, but $(5 - 3) \times 60 = \$120$ loss, the whole fixed cost. The firm needs a subsidy or it leaves.
- **Price = ATC = \$5:** revenue $5 \times 60 = \$300$ covers $\text{TC} = \$300$: zero profit. But $P > \text{MC}$: buyers take fewer connections at \$5 than at \$3, so some deadweight loss remains.

Finding

For a natural monopoly ATC is above MC, so a price equal to MC always makes a loss.

Part 6 summary

- Price discrimination: the same good at different prices. It needs market power and no resale.
- It raises profit and can raise output. Perfect price discrimination reaches $P = \text{MC}$ with no deadweight loss.
- Policy options: antitrust, regulation, public ownership, or doing nothing.
- Regulating a natural monopoly at $P = \text{MC}$ causes a loss; at $P = \text{ATC}$ it breaks even.

Your turn: Check your understanding

Audits: 3 large firms pay up to \$70, 4 small firms up to \$30; each audit costs \$10.

1. With one price, the best profit is:

   A. \$140 B. \$180 C. \$260

2. With two prices, profit rises by:

   A. \$40 B. \$80 C. \$120

3. If buyers can resell, the firm can still charge two prices.

   A. True B. False

Buyers	Number	Most pays
Large firms	3	\$70
Small firms	4	\$30

Solution: Check your understanding

1. **B.** \$70: $3 \times 60 = \$180$. \$30: $7 \times 20 = \$140$. Best: \$180.
2. **B.** $3 \times 60 + 4 \times 20 = \$260$; $260 - 180 = \$80$.
3. **B.** False: small firms would buy at \$30 and resell to large firms.

Buyers	Number	Most pays
Large firms	3	\$70
Small firms	4	\$30

Check: Part 6

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

D can maximize profits if the seller can prevent the resale of goods between customers.

Check: Part 6 – answer

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

D can maximize profits if the seller can prevent the resale of goods between customers.

Finding

Charging each group what it will pay raises profit, as in the audits example. It needs market power (so not competition) and no resale between buyers.

Which rule answers which question

Question	Rule
What demand does a monopoly face?	The market demand curve: $P = P(Q)$.
Why does a monopoly exist?	Barriers to entry: a key resource, government rights, or natural monopoly.
How do P and MR compare?	MR $< P$: the price effect.
How much should it produce?	MR $=$ MC. Increase output if MR $>$ MC; cut it if MC $>$ MR.
What price does it charge?	The price on the demand curve at that output: $P >$ MC.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$; a loss if $P <$ ATC.
How does it compare with competition?	Lower output, higher price, a deadweight loss.